STA2002 Probability and Statistics II
Assignment 2
Due date: 23 October 2023 11:59pm

Please work on the assignment by yourself only. We follow a strict rule on academic honesty.

1. Suppose you flip a coin 314 times, and heads appears 159 times.

   (a) Construct an approximate 90% confidence interval for the probability that the coin comes up heads.

   Let p be the probability that the coin comes up heads. Then $\hat{p} = 159/314$. The 90% confidence interval is

   $$\hat{p} \pm 1.645\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \frac{159}{314} \pm 1.645\sqrt{\frac{159/314 \times (1 - 159/314)}{314}} = [0.450, 0.553].$$

   (b) Approximately how many samples would you need to obtain an approximate 90% confidence interval with width 0.02, while keeping exactly 50.64% of flips appearing heads? The width is

   $$2 \times 1.645\sqrt{\frac{0.5064 \times (1 - 0.5064)}{n}} = 0.02.$$

   Solving for n gives $n = 6763.9541$, and so we need 6764 samples.

   (c) You give the coin to a friend, who also flips the coin 314 times, and obtains an approximate confidence interval [0.390, 0.500] instead. What confidence level did (s)he use?

   The confidence interval is symmetric about the sample mean, which is $(0.39 + 0.5)/2 = 0.445$. The width of the C.I. is

   $$0.5 - 0.39 = 2 \times z_{\alpha/2} \times \sqrt{\frac{0.445(1 - 0.445)}{314}}.$$

   We thus have

   $$z_{\alpha/2} = 1.96,$$

   and so $\alpha = 2(0.025) = 0.05$. The friend constructed an approximate 95% confidence interval.

2. Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu_X, \sigma_X^2)$, $Y_1, \ldots, Y_m \overset{i.i.d.}{\sim} N(\mu_Y, \sigma_Y^2)$ and they are independent. Suppose that both σ_X^2 and σ_Y^2 are known.

   (a) Construct a two-sided $100(1-\alpha)\%$ confidence interval for the difference $\mu_X - \mu_Y$.

   Note that

   $$\overline{X} \sim N\left(\mu_X, \frac{\sigma_X^2}{n}\right),$$
   $$\overline{Y} \sim N\left(\mu_Y, \frac{\sigma_Y^2}{m}\right),$$

   and they are independent, and so $\overline{X} - \overline{Y} \sim N(\mu_X - \mu_Y, \sigma_X^2/n + \sigma_Y^2/m)$. As a result, we have

   $$P\left(-z_{\alpha/2} \leq \frac{(\overline{X} - \overline{Y}) - (\mu_X - \mu_Y)}{\sqrt{\sigma_X^2/n + \sigma_Y^2/m}} \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Rearranging gives the confidence interval to be

$$(\overline{X} - \overline{Y}) \pm z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}.$$

(b) We want to obtain a 90% confidence interval for the difference between true average cable strengths made by Company X and by Company Y. Suppose cable strength is normally distributed for both types of cables with $\sigma_X = 50$ and $\sigma_Y = 30$. If we can make $n + m = 6000$ observations, how many of these should be on Company X cable if we want to minimize the width of the interval?

If we decide to take n samples of Company X cable, the width of the resulting $100(1-\alpha)\%$ confidence interval would be

$$W(n) = 2 \cdot 1.645 \cdot \sqrt{\frac{50^2}{n} + \frac{30^2}{6000 - n}}.$$

Consider the problem of minimizing $W(n)$ with respect to n on the interval $[0, 6000]$. Noting that minimizing $W(n)$ is equivalent to minimizing $W(n)^2$, we instead consider the problem of minimizing

$$f(n) = \big(W(n)\big)^2 = (2 \cdot 1.645)^2 \left(\frac{50^2}{n} + \frac{30^2}{6000 - n}\right).$$

Solving $f'(n) = 0$ yields that the above function is minimized when $n = 3750$. We should thus take 3750 samples of Company X cable, and 2250 samples of Company Y cable. (Notice that these values are proportional to the standard deviations.)

3. Michael owns a bakery. The number of chocolate chips that he adds to his cookies is normally distributed with some mean μ and known variance $\sigma^2 = 121$. A customer buys a dozen of these cookies, and obtains the sample

$$32, 40, 57, 25, 54, 59, 47, 45, 23, 33, 42, 44$$

for which the sample mean is $\bar{x} = 41.75$ and the sample variance is $s^2 \approx 11.8^2$.

(a) Compute a 90% confidence interval for μ.

In lecture, we showed that

$$\left[\bar{x} - 1.645\frac{\sigma}{\sqrt{n}},\ \bar{x} + 1.645\frac{\sigma}{\sqrt{n}}\right].$$

is a 90% confidence interval for μ. Since $\bar{x} = 41.75$, we obtain the interval

$$\left[41.75 - 1.645\frac{11}{\sqrt{12}},\ 41.75 + 1.645\frac{11}{\sqrt{12}}\right] = [36.5,\ 47.0],$$

(b) Compute 95% and 99% confidence intervals for μ.

Since $z_{0.025} = 1.96$ and $z_{0.005} = 2.576$, our 95% confidence interval for μ is

$$\left[41.75 - 1.96\frac{11}{\sqrt{12}},\ 41.75 + 1.96\frac{11}{\sqrt{12}}\right] = [35.5,\ 48.0].$$

and our 99% confidence interval for μ is

$$\left[41.75 - 2.575\frac{11}{\sqrt{12}},\ 41.75 + 2.575\frac{11}{\sqrt{12}}\right] = [33.6,\ 49.9].$$

(c) Repeat part (a), assuming that the customer has no idea what σ^2 is.

$$\left[\bar{x} - t_{0.05}(11)\frac{s}{\sqrt{n}},\ \bar{x} + t_{0.05}(11)\,\frac{s}{\sqrt{n}}\right] = \left[41.75 - 1.796\frac{11.8}{\sqrt{12}},\ 41.75 + 1.796\,\frac{11.8}{\sqrt{12}}\right] = [35.6,\ 47.9].$$

4. Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$. In this question, we will construct confidence interval for σ^2. Let $C \sim \chi^2(r)$, and as usual for any $\alpha \in [0,1]$ we denote $\chi^2_\alpha(r)$ to be

$$P(C > \chi^2_\alpha(r)) = \alpha.$$

(a) Suppose that μ is known. By considering the distribution of

$$\sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma} \right)^2,$$

prove that

$$\left[\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{\alpha/2}(n)}, \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{1-\alpha/2}(n)} \right]$$

is a $100(1-\alpha)\%$ confidence interval for σ^2.

As recalled in Lecture 1 and 2, we have

$$\sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi^2(n),$$

and so

$$P\left(\chi^2_{1-\alpha/2}(n) \leq \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} \leq \chi^2_{\alpha/2}(n) \right) = 1 - \alpha.$$

Rearranging gives

$$P\left(\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{\alpha/2}(n)} \leq \sigma^2 \leq \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{1-\alpha/2}(n)} \right) = 1 - \alpha.$$

(b) Suppose that μ is unknown. By considering the distribution of

$$\sum_{i=1}^{n} \left(\frac{X_i - \overline{X}}{\sigma} \right)^2 = \frac{(n-1)S^2}{\sigma^2},$$

prove that

$$\left[\frac{\sum_{i=1}^n (X_i - \overline{X})^2}{\chi^2_{\alpha/2}(n-1)}, \frac{\sum_{i=1}^n (X_i - \overline{X})^2}{\chi^2_{1-\alpha/2}(n-1)} \right]$$

is a $100(1-\alpha)\%$ confidence interval for σ^2.

As recalled in Lecture 1 and 2, we have

$$\sum_{i=1}^{n} \left(\frac{X_i - \overline{X}}{\sigma} \right)^2 \sim \chi^2(n-1),$$

and so

$$P\left(\chi^2_{1-\alpha/2}(n-1) \leq \frac{\sum_{i=1}^n (X_i - \overline{X})^2}{\sigma^2} \leq \chi^2_{\alpha/2}(n-1) \right) = 1 - \alpha.$$

Rearranging gives

$$P\left(\frac{\sum_{i=1}^n (X_i - \overline{X})^2}{\chi^2_{\alpha/2}(n-1)} \leq \sigma^2 \leq \frac{\sum_{i=1}^n (X_i - \overline{X})^2}{\chi^2_{1-\alpha/2}(n-1)} \right) = 1 - \alpha.$$

(c) In the same setting as part (b), that is, suppose that μ is unknown. Construct a $100(1-\alpha)\%$ confidence interval for σ.

$$P\left(\frac{\sum_{i=1}^n (X_i - \overline{X})^2}{\chi^2_{\alpha/2}(n-1)} \leq \sigma^2 \leq \frac{\sum_{i=1}^n (X_i - \overline{X})^2}{\chi^2_{1-\alpha/2}(n-1)} \right) = 1 - \alpha.$$

$$P\left(\sqrt{\frac{\sum_{i=1}^n (X_i - \overline{X})^2}{\chi^2_{\alpha/2}(n-1)}} \leq \sigma \leq \sqrt{\frac{\sum_{i=1}^n (X_i - \overline{X})^2}{\chi^2_{1-\alpha/2}(n-1)}} \right) = 1 - \alpha.$$

5. Charles suspects that Michael has been adding more blueberries than usual to his muffins. For some reason, he has kept meticulous records, and found that Michael's muffins usually contain a number of blueberries that is normally distributed with mean $\mu_0 = 26$ and standard deviation $\sigma = 4$. However, the last 15 muffins Charles bought contained the following numbers of blueberries:

   21 22 18 29 35 20 30 32 26 23 19 29 31 28 27

   Charles is considerate, and decides that he doesn't want to alarm Michael unless his findings are significant at the $\alpha = 0.05$ level.

   It seems like the design of this problem is questionable

6. If your answer to part (d) is "yes", how small would the sample mean $\overline{x}$ need to be for Charles not to reject H_0? Alternatively, if your answer to part (d) is "no", how large would $\overline{x}$ need to be for Charles to reject H_0?

   Charles rejects the null hypothesis whenever

$$t = \frac{\overline{x} - 26}{4/\sqrt{15}} > z_{0.05} = 1.645 \implies \overline{x} > 1.645\frac{4}{\sqrt{15}} + 26 = 27.593$$

7. Repeat parts (c) and (d), but assume this time that Charles has no idea what σ^2 is.

   This time, our test statistic becomes

$$t = \frac{\overline{x} - 26}{s/\sqrt{15}} = \frac{26.333 - 26}{4.030/\sqrt{15}} \approx 0.32.$$

   and we obtain the p–value

$$p = P(t(14) \geq 0.32) \approx 0.377.$$

(Traffic volume) Michael lives in Minneapolis. He is interested in the traffic volume of Minneapolis. Download the dataset "traffic.csv" on Blackboard. There are 9 columns and 48,204 data points. We will only use two columns for this question: "weather_main" and "traffic_volume". "weather_main" is a short description of the weather, and "traffic_volume" is the hourly traffic volume. Similar to Assignment 1, use your favourite computing language and attach your code at the end of your answer.

Note: the dataset in this question is a real-world dataset from UC Irvine Machine Learning depository. The link is here: CLICK.

1. Michael is interested in doing a comparison between the hourly traffic volume when the weather is "Clear" verus the hourly traffic volume when the weather is "Rain". He has the feeling that there is less traffic when the weather is "Rain", and would like to derive some statistical insights from the data to support his idea. What is the sample mean of the traffic volume when the weather is "Clear", say $\overline{x}$? What is the sample mean of the traffic volume when the weather is "Rain", say $\overline{y}$?

$$\overline{y} = 3317.906, \quad \overline{x} = 3055.909$$

2. Assume the hourly traffic volume when the weather is "Rain" are $Y_1, \ldots, Y_m \overset{i.i.d.}{\sim} N(\mu_Y, \sigma^2)$, and the hourly traffic volume when the weather is "Clear" are $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu_X, \sigma^2)$, and X_i and Y_i are independent. These assumptions are not realistic but still provide us a model to work on. Compute the 95% two-sample pooled t-interval for the difference $\mu_X - \mu_Y$. As n and m are large in this dataset, please use $z_{0.025}$ to replace $t_{0.025}(n + m - 2)$ when you compute the interval.

$$
\begin{aligned}
n &= 13391 \\
m &= 5672 \\
s_p^2 &= \frac{\sum_{i=1}^n (x_i - \overline{x})^2 + \sum_{i=1}^m (y_i - \overline{y})^2}{n+m-2} = 3942817.599 \\
\overline{x} - \overline{y} \pm 1.96 s_p \sqrt{\frac{1}{n} + \frac{1}{m}} &= 3055.909 - 3317.906 \pm 1.96\sqrt{3942817.599}\sqrt{1/13391 + 1/5672} \\
&= [-323.653, -200.340].
\end{aligned}
$$

3. Assume the hourly traffic volume when the weather is "Rain" are $Y_1, \ldots, Y_m \overset{i.i.d.}{\sim} N(\mu_Y, \sigma_Y^2)$, and the hourly traffic volume when the weather is "Clear" are $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu_X, \sigma_X^2)$, and X_i and Y_i are independent and $\sigma_X^2 \neq \sigma_Y^2$. Compute the 95% Welch's t-interval for the difference $\mu_X - \mu_Y$. As n and m are large in this dataset, please use $z_{0.025}$ to replace the t distribution quantiles when you compute the interval.

$$
\begin{aligned}
s_x^2 &= 3948572.016 \\
s_y^2 &= 3929230.639 \\
\overline{x} - \overline{y} \pm 1.96\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}} &= [-323.592, -200.401].
\end{aligned}
$$

(Traffic volume continued) Recall the last question of Assignment 2. Michael lives in Minneapolis. He is interested in the traffic volume of Minneapolis. Download the dataset "traffic.csv" on Blackboard. There are 9 columns and 48,204 data points. We will only use two columns for this question: "weather_main" and "traffic_volume". "weather_main" is a short description of the weather, and "traffic_volume" is the hourly traffic volume. Use your favourite computing language and attach your code at the end of your answer.

Assume the hourly traffic volume when the weather is "Rain" are $Y_1, \ldots, Y_m \overset{i.i.d.}{\sim} N(\mu_Y, \sigma_Y^2)$, and the hourly traffic volume when the weather is "Clear" are $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu_X, \sigma_X^2)$, and X_i and Y_i are independent and $\sigma_X^2 \neq \sigma_Y^2$.

1. Michael is interested in doing a comparison between the hourly traffic volume when the weather is "Clear" verus the hourly traffic volume when the weather is "Rain". He has the feeling that there is less traffic on average when the weather is "Rain", and would like to derive some statistical insights from the data to support his idea. Formulate the (one-sided) null hypothesis H_0 and the alternative hypothesis H_1.

$$H_0 : \mu_X = \mu_Y, \quad H_1 : \mu_X > \mu_Y$$

2. Assume that $\sigma_X^2 = \sigma_Y^2$. Compute the test statistic and conduct an appropriate hypothesis test at 5% significance level.

   We conduct the pooled t-test. The test statistic is

$$t = \frac{3055.909 - 3317.906}{\sqrt{3942817.599}\sqrt{1/13391 + 1/5672}} = -8.3286.$$

   Since

$$t = -8.3286 < t_{0.05}(19061) \approx z_{0.05} = 1.645,$$

   we fail to reject H_0 at 5% significance level.

3. Assume that $\sigma_X^2 \neq \sigma_Y^2$. Compute the test statistic and conduct an appropriate hypothesis test at 5% significance level.

   We conduct the Welch's t-test. The test statistic is

$$t = \frac{3055.909 - 3317.906}{\sqrt{\frac{3948572.016}{13391} + \frac{3929230.639}{5672}}} = -8.33688.$$

   Since

$$t = -8.33688 < t_{0.05}(r) \approx z_{0.05} = 1.645,$$

   we fail to reject H_0 at 5% significance level.

4. Conduct the F-test to test

$$H_0 : \sigma_X^2 = \sigma_Y^2, \quad H_1 : \sigma_X^2 \neq \sigma_Y^2.$$

   at 5% significance level.

$$\begin{aligned} n &= 13391 \\ m &= 5672 \\ s_x^2 &= 3948572.016 \\ s_y^2 &= 3929230.639 \\ t &= \frac{s_x^2}{s_y^2} = 1.004922 \\ F_{0.025}(13391-1, 5672-1) &= 1.0451 \\ F_{0.975}(13391-1, 5672-1) &= 0.957 \end{aligned}$$

   Since $F_{0.975}(13391-1, 5672-1) < t < F_{0.025}(13391-1, 5672-1)$, we fail to reject H_0 at 5% significance level.